

Design and Deploy a Real-World AI Agent

Real-World Agent

This capstone brings together everything you've learned across the course. Your task is to design, build, test, and document an intelligent agent capable of performing a multi-step task autonomously in a dynamic environment. Your agent should use structured perception, decision logic, memory, and tool integration—and be resilient to ambiguity, failure, and change.

Note: Agent use case category: e.g., customer support, task planning

This is your opportunity to prove you can not only build a functional agent—but engineer one that works reliably and intelligently, even when the real world gets messy.

Instructions for learners

Project Requirements

Your agent must:

→ Be goal-driven (designed to complete a specific real-world task)

→ Use multi-step planning or decision logic (not just a one-turn response)

→ Integrate at least one external tool or API (search, database, scheduler, etc.)

→ Include a memory component (short-term or persistent)

→ Handle at least one failure or fallback case (e.g., tool error, bad input, ambiguous intent)

Choose your own, or modify one of the below:

- An email assistant that drafts, summarizes, and tracks follow-ups
- A research bot that explores a topic, compiles insights, and stores structured notes
- A support triage agent that collects symptoms, checks a knowledge base, and escalates if unresolved
- A task automation agent that pulls inputs from multiple systems and returns an organized output

What to Submit

→ Your final submission must include:

→ Agent Overview (2–3 paragraphs):

→ Describe the problem it solves

→ Explain its target users

→ Outline how it operates

Architecture Design

- Specify whether the agent is reactive, deliberative, or hybrid
- Diagram or list its key modules: perception, planning, tools, memory, fallback

Code or Logic Snippet

- Submit a meaningful excerpt (Long Chain chain, Rasa flow, planner logic, etc.)
- Highlight a part that handles context, tool orchestration, or error recovery

Behavior Trace or Walkthrough

- Provide 1 example of a real interaction (user input → decisions → result)
- Include 1 example of how your agent handles a failure or edge case

Reflection

Reflection (3–5 sentences)

- What's one design decision you're most proud of?
- What was the hardest part to get right?
- How would you improve this agent in a second iteration?

Completion

Goals

By completing this capstone, you'll demonstrate your ability to:

Design agent systems that combine perception, logic, tools, and memory

Debug and refine agents for performance and resilience

Document agent behavior in a way that shows transparency and reasoning

Apply real-world design thinking to intelligent agent development